

Unlocking Success:

Identifying Key Factors Driving Job Seekers' Outcomes

SKILLUP X TEAM TGW

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Introduction -Company & case

SKILLUP

Nonprofit organization founded in 2020, helps workers without a degree secure quality jobs through career navigation, training, and resources, empowering them to develop skills and enter high-demand industries.

About Case

Using SkillUp website data and survey to:

- **Track user behaviors** – Analyze user interactions, including resource access, coaching, and job applications.
- **Identify key success factors** – Identify key drivers of job placement and training participation.
- **Understand demographic variations** – Examine how success factors differ across user backgrounds and career goals.

Methodologies & Analysis -Overview

1. Data Cleaning Process

- **Confirm Data Sources:** WAP_pulse_survey, WAP_all_time_connect, WAP_individual
- **Define Analysis Period:** Q1:2023-Q3:2024
- **Define X and Y Variables:** Y: skillup_find_new_job, skillup_training/X: user interactions, demographics, and engagement metrics
- **Handle Missing Data:** Apply appropriate imputation strategies (category "Unknown" for categorical variables)

2. Data Analysis

- **Descriptive Analysis:** Correlation analysis, Chi-Square Test
- **Prediction Model:** Logistic Regression, Random Forest

Methodologies & Analysis -Data Cleaning Process

WAP_pulse_survey

- 18,696 rows (delete duplicate)
- X='desired_industry'
- Y='skillup_training', 'skillup_find_new_job'

WAP_id,
survey_date>=createdAt+180

WAP_individual

- 18,696 rows
- X='rural_type','myPathCompleted','preferredRegion','preferredRegionGroup','educationAndWorkExperience','workExperienceGroup','race','highestEducationLevel','latestSalary','targetSalary','completed_career_milestone','completed_jobs_milestone','completed_training_milestone','gender_3','age'

WAP_id

WAP_all_time_connect

- X=
 - 1) How many times did the seeker connect to the website in the 180 days before the survey? (e.g. 'Website Visits (Last 180 Days)')
 - 2) Did seekers engage in specific actions on the platform within the 180 days? (e.g. 'Training Clicked Apply')
(Combine 'collectionType' and 'eventName' to identify seekers' behaviors)

Methodologies & Analysis -Data Analysis

Descriptive Analysis: Chi-Square Test

Outcome: SkillUp Training

Did you find the training program through SkillUp?

Skillup Training	Count
No	17,579
Yes	1,117

Variable	Chi2	p-value
Race	115.5999956	0
Gender	75.57293787	0
Education And Work Experience	61.66465954	0
Preferred Region	59.01171643	0.0005
Desired Industry	41.20101301	0
Work Experience Group	35.33558978	0
Highest Education Level	30.63702867	0
Completed Training Milestone	23.04511922	0
Completed Jobs Milestone	18.10487572	0
Training Clicked Apply	13.42990587	0.0002
Jobs Clicked Apply	2.57384253	0.1086
Rural Type	2.356067871	0.1248
Resources Saved Catalog Item	2.217135956	0.1365
Careers Viewed Catalog Item	2.193022305	0.1386
Training Saved Catalog Item	2.008890886	0.1564
Training Edited Status	1.319843171	0.2506
Jobs Edited Status	0.780333306	0.377
Completed Career Milestone	0.582221076	0.4454
Jobs Viewed Catalog Item	0.381716748	0.5367
Training Viewed Catalog Item	0.339418452	0.5602
Preferred Region Group	0.193305569	0.6602
Careers Saved Catalog Item	0.003714076	0.9514
Jobs Saved Catalog Item	0	1
Missing Edited Status	0	1
Resources Clicked on a Resource	0	1

Outcome: SkillUp Find New Job

Did you find your new job through SkillUp?

Skillup Find New Job	count
FALSE	18,122
TRUE	574

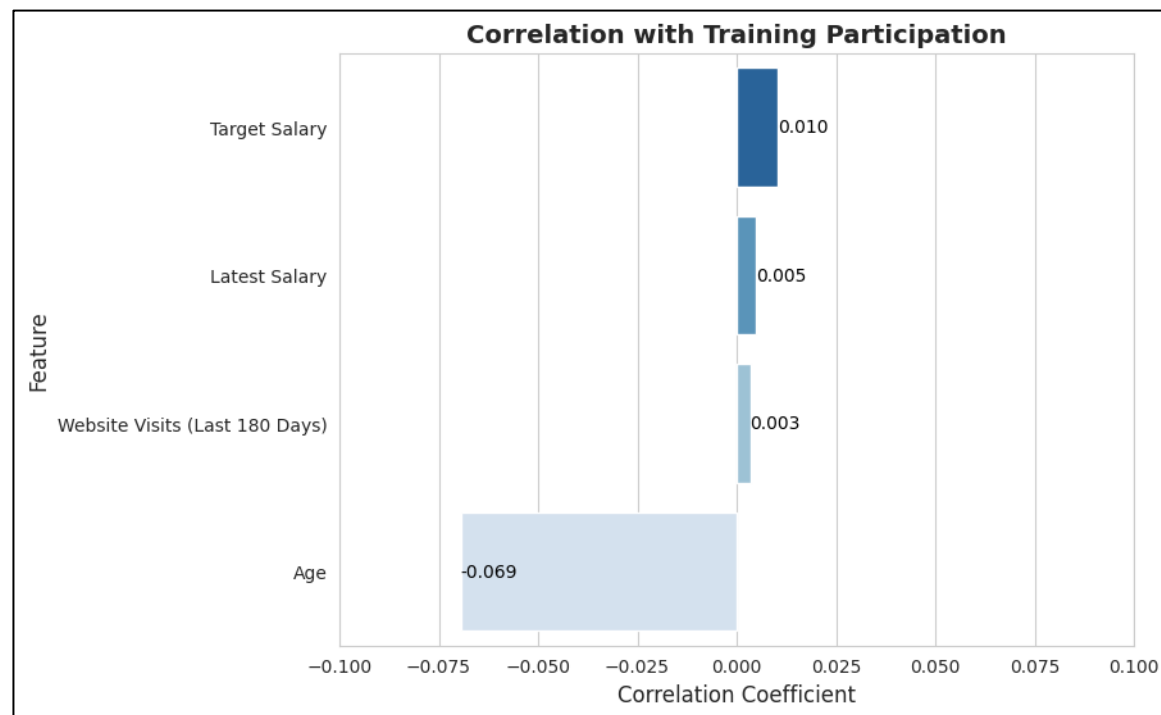
Variable	Chi2	p-value
Gender	130.0592522	0
Race	68.68995172	0
Highest Education Level	64.90057847	0
Education And Work Experience	56.42146185	0
Preferred Region	46.59777643	0.0151
Desired Industry	44.78321045	0
Jobs Clicked Apply	7.685240464	0.0056
Completed Training Milestone	7.453409661	0.0063
Rural Type	5.148964612	0.0233
Careers Viewed Catalog Item	5.107950227	0.0238
Training Saved Catalog Item	5.045713077	0.0247
Preferred Region Group	1.333096149	0.2483
Careers Saved Catalog Item	1.28851688	0.2563
Training Clicked Apply	1.153477156	0.2828
Completed Career Milestone	1.130675683	0.2876
Training Viewed Catalog Item	0.872392513	0.3503
Jobs Saved Catalog Item	0.364405466	0.5461
Resources Saved Catalog Item	0.236935804	0.6264
Training Edited Status	0.171996328	0.6783
Jobs Viewed Catalog Item	0.078501475	0.7793
Work Experience Group	0.011733094	0.9137
Resources Clicked on a Resource	0	1
Missing Edited Status	0	1
Completed Jobs Milestone	0	1
Jobs Edited Status	0	1

Methodologies & Analysis -Data Analysis

Descriptive Analysis: Correlation analysis

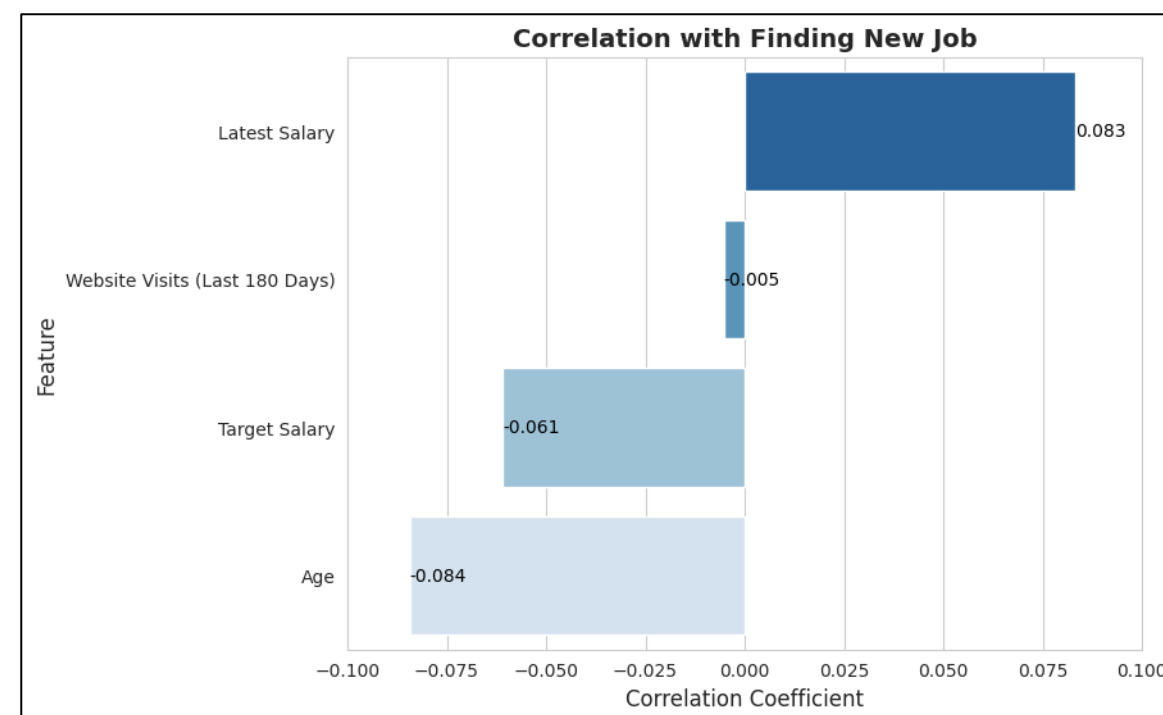
❑ Outcome: SkillUp Training

Did you find the training program through SkillUp?



❑ Outcome: SkillUp Find New Job

Did you find your new job through SkillUp?



Methodologies & Analysis -Data Analysis

Prediction Model: Logistic Regression

Outcome: SkillUp Training

```
Logistic Regression Performance
              precision    recall  f1-score   support

   False      0.63      0.50      0.56      347
    True      0.56      0.68      0.61      324

 accuracy      0.59      0.59      0.59      671
 macro avg      0.59      0.59      0.59      671
weighted avg      0.60      0.59      0.59      671

Top 10 Most Important Features in Logistic Regression
Desired Industry_Skilled Trades      0.000303
Gender_Other, declined      0.000347
Preferred Region_New York City      0.000445
Training Clicked Apply_1.0      0.000500
Highest Education Level_High school diploma or equivalent      0.000547
Education And Work Experience_2-4 years      0.000608
Gender_Male      0.000678
Desired Industry_Technology      0.000780
Race_Black or African American      0.000928
Work Experience Group_<5 years      0.000954
dtype: float64
```

Outcome: SkillUp Find New Job

```
Logistic Regression Performance
              precision    recall  f1-score   support

   False      0.73      0.66      0.69      184
    True      0.65      0.73      0.69      161

 accuracy      0.69      0.69      0.69      345
 macro avg      0.69      0.69      0.69      345
weighted avg      0.69      0.69      0.69      345

Top 10 Most Important Features in Logistic Regression
Highest Education Level_High school diploma or equivalent      0.520520
Preferred Region_Chicago      0.528942
Preferred Region_New York City      0.561208
Education And Work Experience_2-4 years      0.581675
Gender_Male      0.619004
Preferred Region_Denver      0.622288
Race_Middle Eastern or North African      0.631632
Training Clicked Apply_1.0      0.666073
Completed Training Milestone_True      0.720353
Desired Industry_Supply Chain & Logistics      0.892627
dtype: float64
```


Methodologies & Analysis -Data Analysis

Prediction Model: Random Forest

Outcome: SkillUp Training

Random Forest Performance				
	precision	recall	f1-score	support
False	0.61	0.57	0.59	347
True	0.57	0.60	0.59	324
accuracy			0.59	671
macro avg	0.59	0.59	0.59	671
weighted avg	0.59	0.59	0.59	671
Top 10 Most Important Features in Random Forest				
Age				0.191642
Website Visits (Last 180 Days)				0.052761
Desired Industry_Technology				0.027887
Race_White				0.027674
Desired Industry_Healthcare				0.026821
Gender_Male				0.025556
Rural Type_Rural				0.025190
Highest Education Level_High school diploma or equivalent				0.024311
Race_Black or African American				0.023106
latestSalary				0.022295
dtype: float64				

Outcome: SkillUp Find New Job

Random Forest Performance				
	precision	recall	f1-score	support
False	0.68	0.61	0.64	184
True	0.60	0.68	0.64	161
accuracy			0.64	345
macro avg	0.64	0.64	0.64	345
weighted avg	0.65	0.64	0.64	345
Top 10 Most Important Features in Random Forest				
Age				0.188075
Website Visits (Last 180 Days)				0.045649
Gender_Male				0.033985
Highest Education Level_High school diploma or equivalent				0.033685
Desired Industry_Technology				0.030632
Race_Black or African American				0.028705
Desired Industry_Healthcare				0.027079
Education And Work Experience_5-10 years				0.023819
Highest Education Level_Some college				0.023656
Race_White				0.023184
dtype: float64				

Key findings & insights

Identify key success factors

▣ SkillUp Training Participation

- **Demographics:** Age, Race, Gender
- **Education & Work Background:** Highest Education Level, Work Experience
- **Geographical Influence:** Preferred Region
- **Career & Industry Factors:** Desired Industry,
- **Engagement :** Completed Training Milestone, Completed Jobs Milestone, Training Clicked Apply

▣ Finding a New Job through SkillUp

- **Demographics:** Age, Race, Gender
- **Education & Work Background:** Highest Education Level, Work Experience, Latest Salary
- **Geographical Influence:** Preferred Region, Rural Type
- **Career & Industry Factors:** Desired Industry
- **Engagement :** Completed Training Milestone, Jobs Clicked Apply, Careers Viewed Catalog Item, Training Saved Catalog Item



✓ **Demographics & Career Milestones Matter** – Age, race, gender, and completing key milestones (e.g., training, job applications) impact both training and job success.

✓ **Regional & Industry Preferences Influence Outcomes** – Users in specific regions and industries (e.g., Technology, Healthcare) show higher training and job placement rates.

✓ **Platform Engagement is Key** – Actions like applying for training, saving jobs, and interacting with career content boost both participation and job success.

Key findings & insights

Does Race Impact Career Milestones?

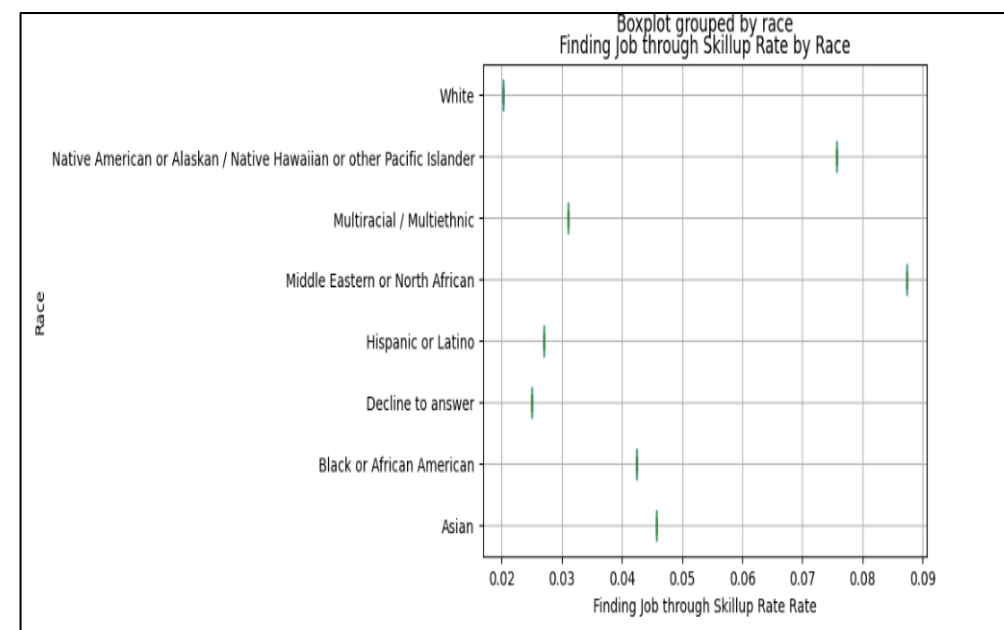
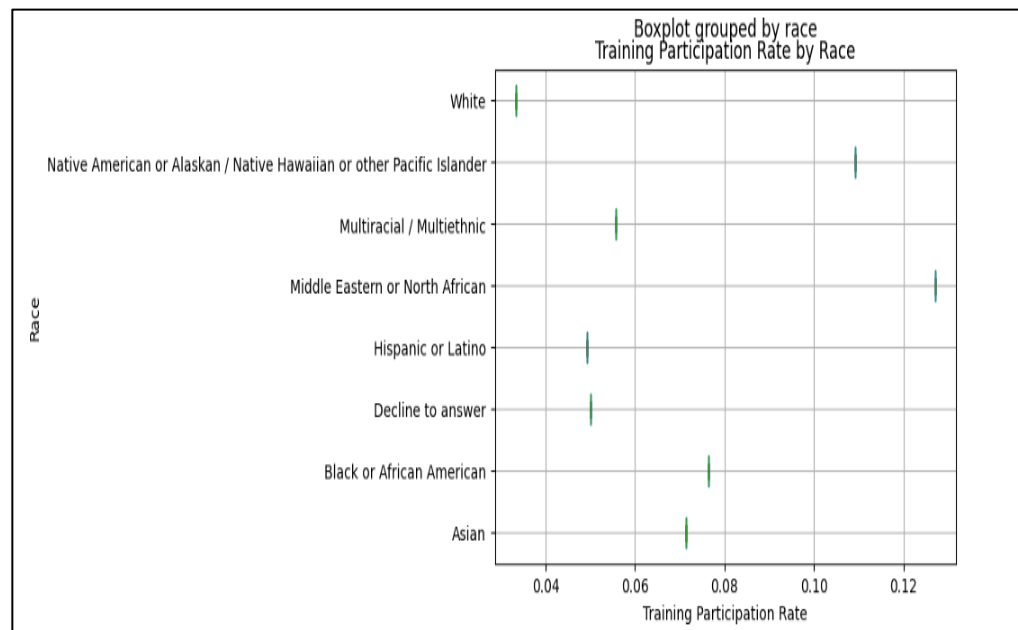


Training Participation Rate:

Middle Eastern or North African (12.7%) and Native American or Alaskan / Native Hawaiian (10.9%) have the highest training participation rates. White individuals (3.3%) have the lowest participation rate.

Finding a Job Through Skillup:

Middle Eastern or North African (8.7%) and Native American or Alaskan / Native Hawaiian (7.6%) have the highest job finding rates. White individuals (2.0%) have the lowest job finding rate.



Key findings & insights

Does Race Impact Career Milestones?

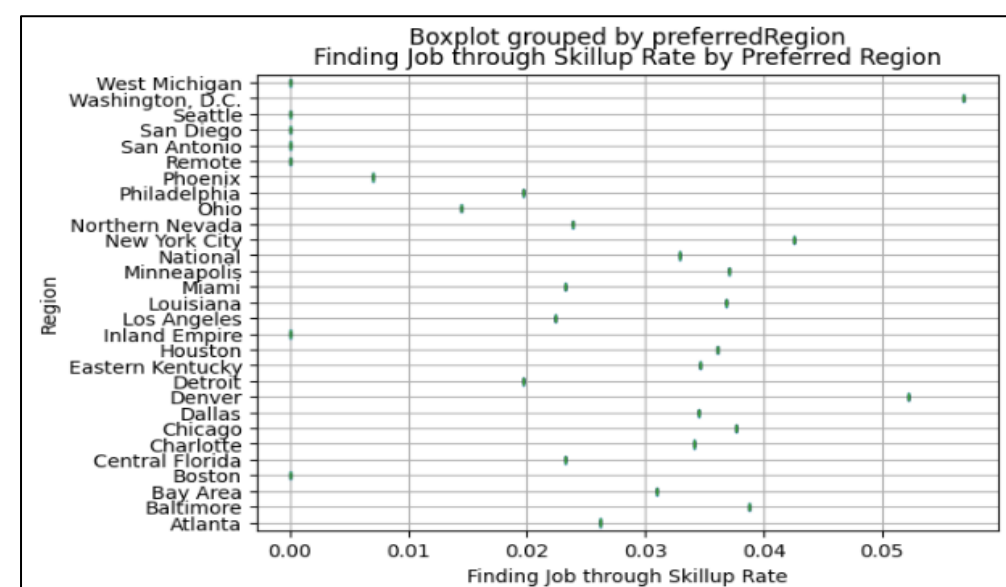
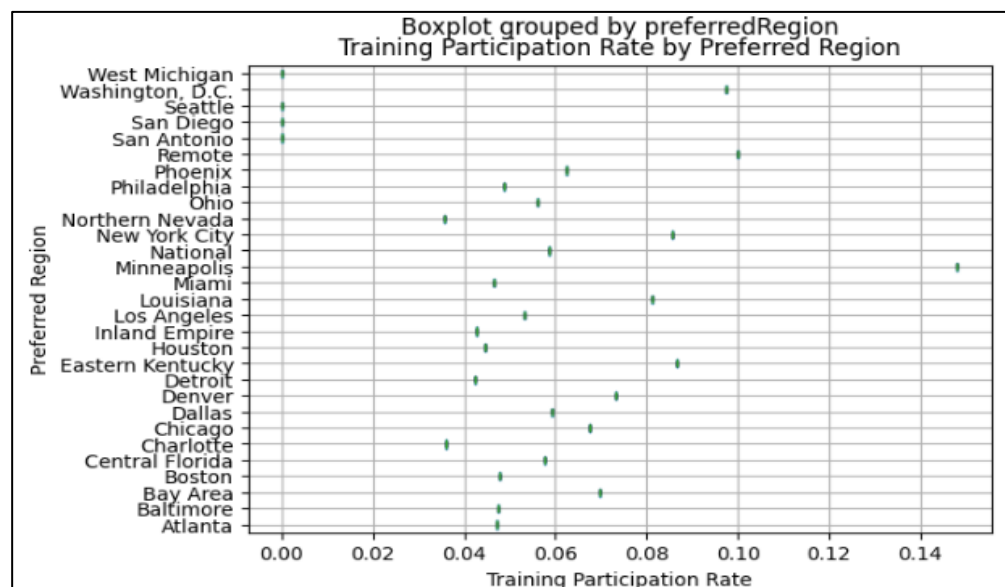


Training Participation Rate:

Training participation varies significantly based on the seeker's desired job location. Washington, D.C. has a higher training participation rate, whereas locations like Seattle show a 0% participation rate, indicating no recorded engagement from seekers targeting those regions.

Finding a Job Through Skillup:

Job attainment through SkillUp also differs by the seeker's preferred job location. Some locations, such as Denver and Washington, D.C., have higher job placement rates, while others, like Seattle, show a 0% job placement rate, indicating no recorded job outcomes for seekers aiming to work there.



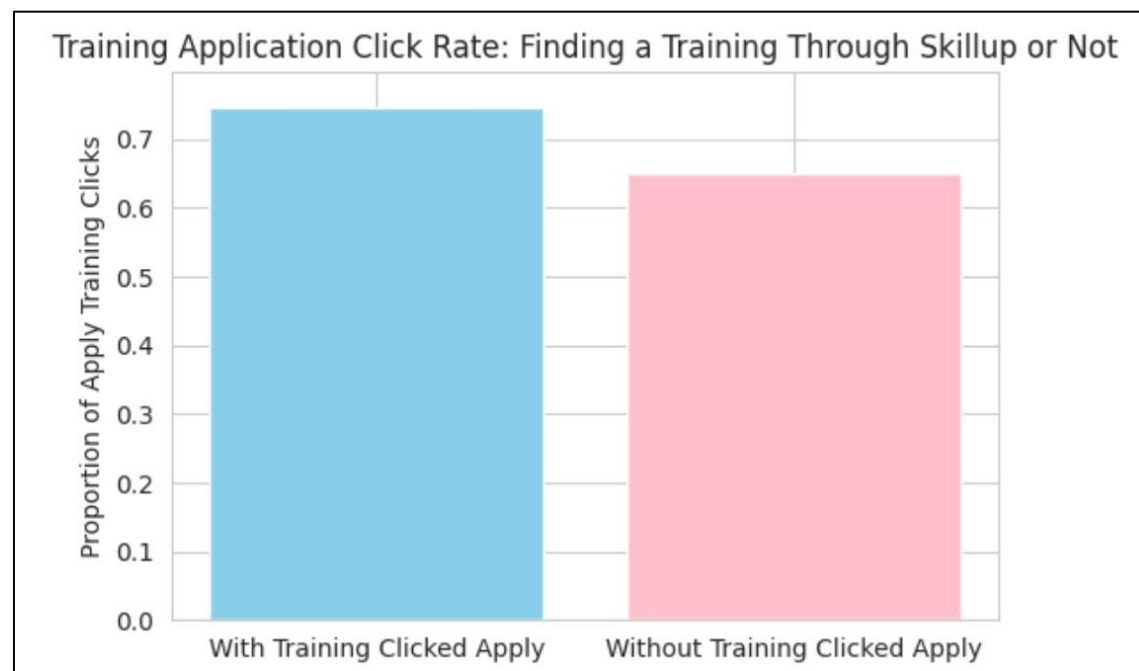
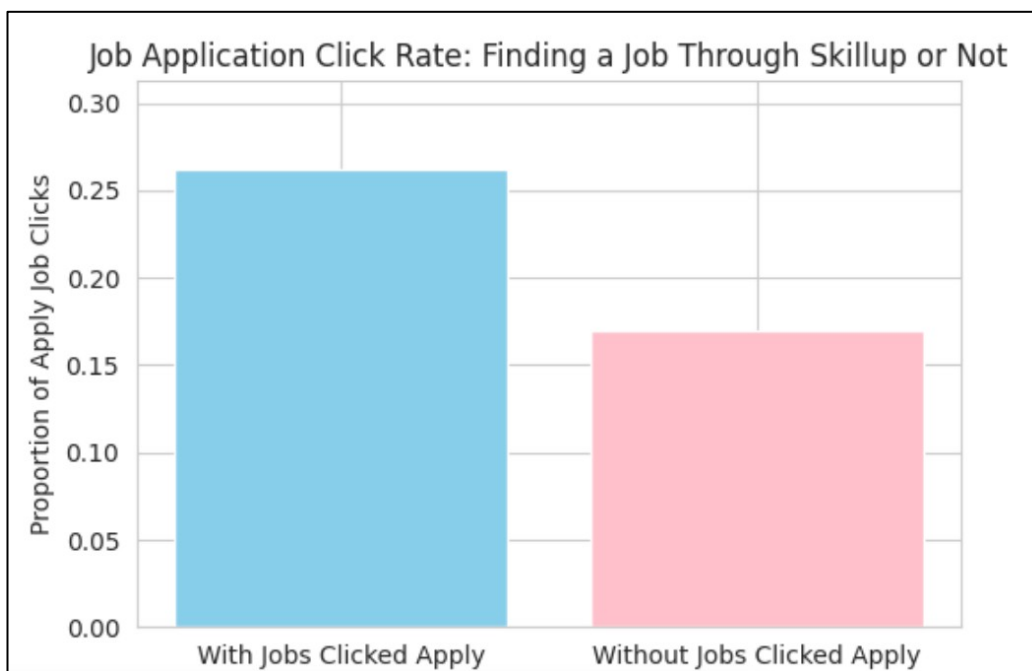
Key findings & insights

Typical User Journey to Success



Job and Training Application Activity Correlates with Success

Individuals who successfully found a job through SkillUp had a higher likelihood of engaging in job applications on the platform. Similarly, those who enrolled in training programs had a higher rate of training application clicks. This highlights the importance of active engagement in the application process for achieving career advancement.



Recommendations



Optimize User Engagement on the Platform :

Since both the frequency of website visits and specific interactions are key predictors of training participation and job success, SkillUp should implement features that encourage consistent and meaningful engagement..

- Implement **reminder notifications** for users who frequently visit the site but have not yet applied for training or jobs.
- Provide a **clear career progression roadmap** for users, showing how different training and job applications contribute to their long-term goals.



Consider Regional and Industry-Specific Strategies :

- Tailor SkillUp programs to align with the unique demands of different regions and industries to maximize effectiveness.

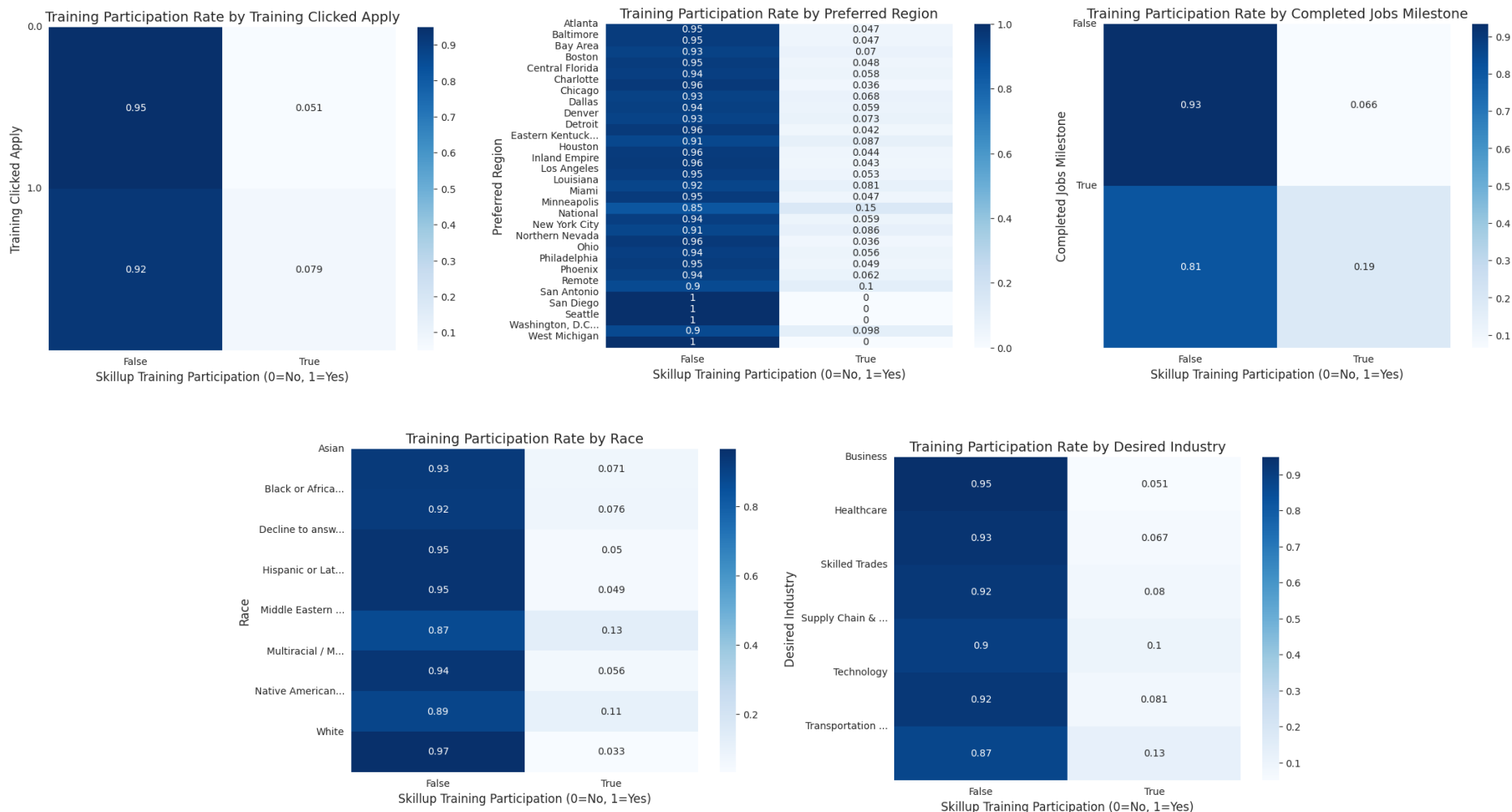


Targeted Training and Employment Support for High-Success Industries :

- Given the higher job success rates in Technology and Healthcare, prioritizing skill development and career advancement initiatives in these sectors will yield better outcomes.

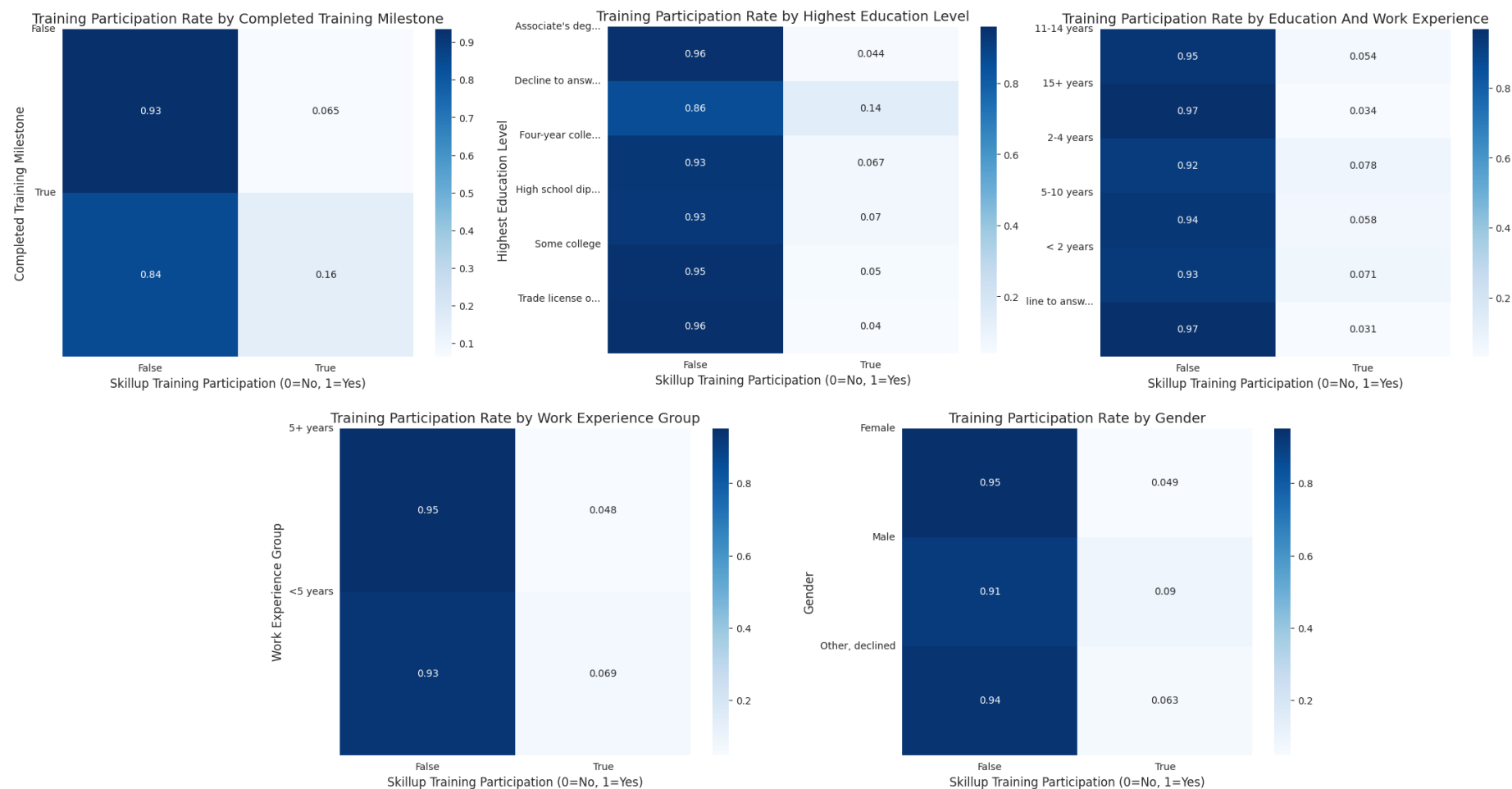
Appendix-Data Analysis

Descriptive Analysis: Heatmap-Analyzing Significant Factors Affecting Training Participation



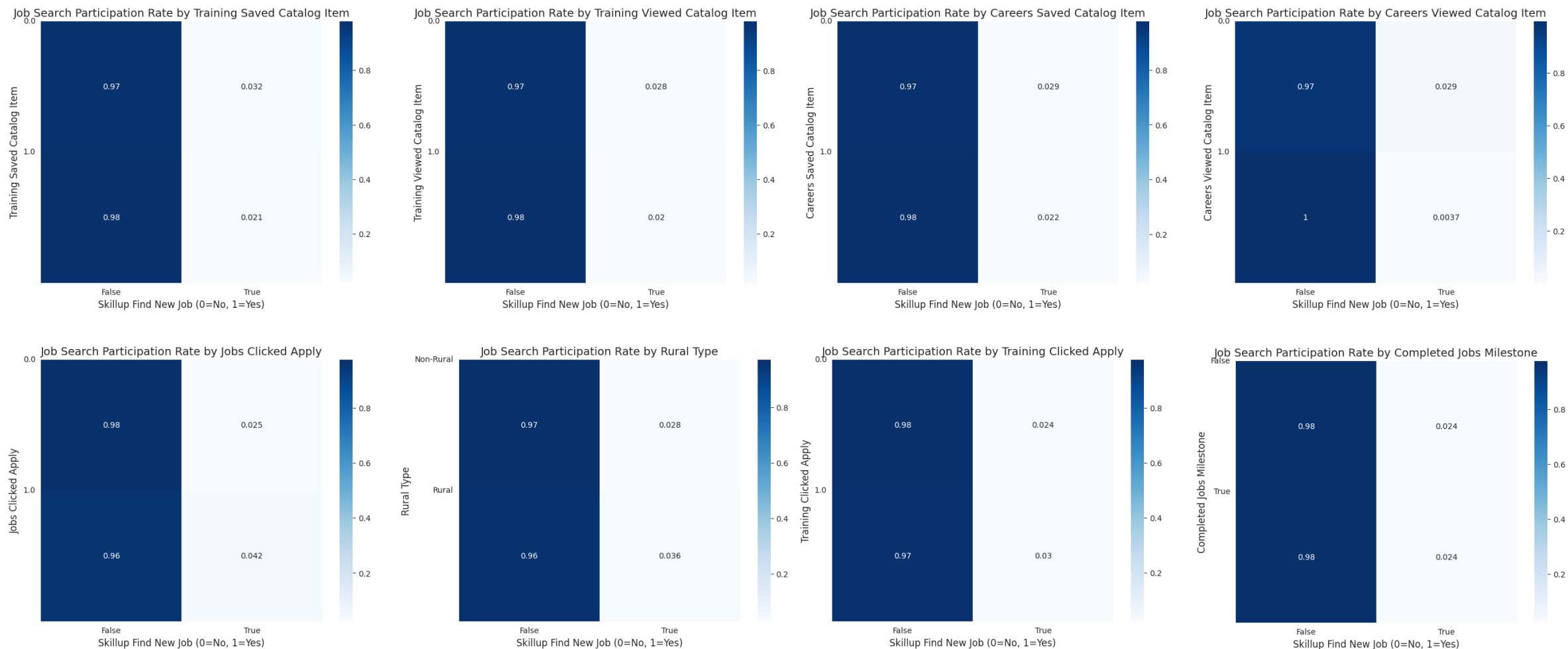
Appendix-Data Analysis

Descriptive Analysis: Heatmap-Analyzing Significant Factors Affecting Training Participation



Appendix-Data Analysis

Descriptive Analysis: Heatmap-Analyzing Significant Factors Affecting Finding New Job Rate



Appendix-Data Analysis

Descriptive Analysis: Heatmap-Analyzing Significant Factors Affecting Finding New Job Rate

